

Adventist University of Central Africa

Information Technology

Software Engineering

Case Flow

A Student Disciplinary Case Management System

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Requirements for the Degree of

Bachelor of Science in Information Technology

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Figure 1: System Architecture

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ABSTRACT

CaseFlow is a web-based Student Disciplinary Case Management System designed to digitize and streamline the entire student disciplinary process at the Adventist University of Central Africa (AUCA), from incident reporting through to re-integration. The system addresses persistent challenges in the current manual process, including slow committee review timelines, lack of transparency for students and staff, inconsistent enforcement of disciplinary decisions, and unreliable records of student re-integration after a disciplinary period. The name “CaseFlow” was deliberately chosen to convey a calm, non-threatening tone appropriate to a system dealing with sensitive student matters, while emphasizing the smooth, structured flow of a case from reporting to resolution.

CaseFlow covers the complete disciplinary case lifecycle: incident reporting by staff or students, secure online review and deliberation by the disciplinary committee, enforcement of decisions tied directly to a student’s registration status, real-time student status tracking, a structured appeals process, and verified re-integration once a sanction period ends. By replacing paper-based and ad-hoc communication with a centralized digital platform, CaseFlow aims to reduce case resolution time, improve transparency and accountability, maintain accurate historical records, and ensure that students re-entering academic life after a disciplinary case are properly verified and supported.

The system will be developed using an Agile methodology with iterative sprints, role-based access control to protect sensitive case information, and a relational database to maintain a complete, auditable history of every case. Expected outcomes include faster committee turnaround times, improved trust in the disciplinary process among students and staff, fewer lost or misplaced case records, and a reliable mechanism for confirming a student’s re-integration status before resuming normal academic activity.

INTRODUCTION

Background

Student discipline is a core part of maintaining a safe, orderly, and value-driven academic environment at any university. At AUCA, as in many institutions, the disciplinary process involves several stages: an incident is reported, a committee reviews the case and gathers information, a decision is made, a sanction (if any) is enforced, and eventually the student may be re-integrated into normal academic life. Each of these stages currently depends heavily on manual coordination — paper forms, in-person meetings, email threads, and physical files stored in administrative offices.

As university populations grow and disciplinary cases become more frequent and varied, manual handling of these processes becomes increasingly difficult to manage consistently. Delays in committee scheduling, loss or misplacement of physical case files, and unclear communication between the committee, the affected student, and academic departments are common consequences of a process that has not kept pace with the digital tools already used elsewhere in the university (such as registration and results systems).

CaseFlow proposes to bring the same level of digital efficiency already present in academic registration to the disciplinary process. By digitizing incident reporting, committee deliberation, enforcement, and re-integration verification, the system aims to make the disciplinary process faster, fairer, and easier to track for everyone involved — students, committee members, and academic administration.

Problem Statement

The current student disciplinary process at AUCA is managed largely through manual, paper-based, and informal communication channels. This creates several recurring problems. First, disciplinary committee timelines are slow and unpredictable, since scheduling reviews and circulating case information depends on the physical or informal availability of committee members. Second, there is little transparency for the student involved in a case — students often do not know the status of their case, what evidence or testimony has been considered, or when a decision will be made. Third, enforcement of disciplinary decisions is inconsistently linked to a student's registration status, meaning that a student under sanction may continue to register for courses or access services they should not, due to gaps in communication between the disciplinary committee and the registration office.

Fourth, and perhaps most critically, re-integration records are unreliable. When a student completes a sanction period and is expected to resume normal academic standing, there is frequently no clear, verifiable record confirming that re-integration has been properly authorized, leaving both the student and academic staff uncertain about the student's current status. This lack of a single source of truth results in delays, disputes, lost paperwork, and reduced confidence in the fairness and consistency of the disciplinary system.

Without a centralized, transparent, and auditable system to manage the full disciplinary case lifecycle, the university risks inconsistent enforcement, prolonged case resolution times, frustrated students and staff, and incomplete institutional records of disciplinary history.

Project Justification

Addressing this problem is important because a well-managed disciplinary process is central to institutional integrity, student welfare, and fair treatment. Delays and inconsistencies in handling disciplinary cases can damage trust between students and the institution, while poor record-keeping creates legal and administrative risk for the university. A digital case management system directly supports the university's broader move toward digitizing administrative processes, mirroring efforts already seen in academic registration and result management.

CaseFlow is justified by the tangible benefits it offers: faster committee review cycles through structured online workflows, increased transparency for students through real-time status visibility, reliable enforcement through direct integration with registration status, a clear and fair appeals mechanism, and verifiable re-integration records that protect both the student and the institution. These benefits align with good governance practices in higher education and contribute to a campus environment where disciplinary matters are handled efficiently, consistently, and respectfully.

Objectives

General Objective

To design and develop CaseFlow, a web-based Student Disciplinary Case Management System that digitizes and streamlines the full disciplinary case lifecycle — from incident reporting to re-integration — in order to improve transparency, consistency, and efficiency in handling student disciplinary matters at AUCA.

Specific Objectives

1. To design and implement an incident reporting module that allows authorized staff to securely submit and document disciplinary incidents, including supporting details and evidence.

2. To develop an online committee review module that enables the disciplinary committee to access case information, deliberate, record decisions, and manage cases collaboratively without requiring physical meetings for every stage.
3. To implement an enforcement mechanism that links disciplinary decisions directly to a student's registration status, ensuring sanctions are automatically reflected and enforced within the academic system.
4. To build a student status tracking feature that gives students and authorized staff real-time visibility into the current stage and status of a disciplinary case.
5. To design a structured appeals workflow that allows students to formally contest a decision and have it reviewed through a clear, recorded process.
6. To develop a re-integration verification module that confirms and records when a student has completed a sanction and has been formally cleared to resume normal academic standing.

Scope

The scope of this project defines what CaseFlow will and will not include, ensuring the system remains focused, feasible, and aligned with the disciplinary process at AUCA.

The project will include:

- Incident reporting: A secure module for authorized staff to log disciplinary incidents with relevant details, categories, and supporting documentation.
- Online committee review: A digital workspace where committee members can view case details, add notes, deliberate, and record formal decisions.
- Enforcement tied to registration: Automatic flagging or restriction of a student's registration status based on active disciplinary sanctions.
- Student status tracking: A dashboard where students and staff can view the current stage of a case (reported, under review, decided, under appeal, re-integrated).
- Appeals management: A workflow allowing students to submit appeals against decisions, with committee review and resolution tracking.
- Re-integration verification: A formal process and record confirming a student's return to normal academic standing after a sanction period.
- Role-based access control: Distinct permissions for students, reporting staff, committee members, and administrators to protect sensitive case data.
- Case history and audit trail: A complete, timestamped record of all actions taken on a case for accountability and reference.
- Notifications: Automated alerts to relevant parties (student, committee, registration office) at key stages of a case.

The project will not include:

- Legal representation or advisory services: CaseFlow manages process and records; it does not provide legal counsel to students.
- Criminal justice integration: The system is limited to internal university disciplinary matters and does not interface with external law enforcement or judicial systems.
- Counseling or psychological support services: While the system tracks case status, it does not replace dedicated student counseling services.
- Full HR or staff disciplinary management: The system is scoped specifically to student disciplinary cases, not staff conduct processes.

LITERATURE REVIEW

Introduction

Case management systems have been widely adopted across sectors — legal, healthcare, customer service, and human resources — to bring structure, accountability, and traceability to processes that were traditionally handled manually. In higher education, disciplinary case management remains an area where many institutions still rely on manual or semi-digital processes, despite the clear benefits demonstrated by digital case management in other domains. This literature review examines existing approaches to case and workflow management systems, their application in educational and administrative contexts, and the gaps that CaseFlow aims to address.

Review of Existing Works

Workflow and Case Management Systems

General-purpose case management and workflow systems, such as those used in legal practice management and customer support ticketing, demonstrate the value of structured case lifecycles — intake, assignment, review, decision, and closure — supported by role-based access and audit trails. These systems consistently show that digitizing case handling reduces resolution time and improves accountability compared to manual, paper-based tracking.

Student Conduct and Disciplinary Systems in Higher Education

A number of universities, particularly in North America, have adopted dedicated student conduct management platforms to handle disciplinary referrals, hearings, and sanction tracking. These systems typically centralize incident reports, automate notifications to relevant offices, and maintain a searchable history of student conduct records. However, such systems are often costly, proprietary, and designed around the regulatory and procedural context of specific countries, making them difficult to adapt directly to the policies and processes of institutions such as AUCA.

Integration of Disciplinary Status with Academic Records

Literature on student information systems highlights the importance of linking disciplinary status with registration and enrollment records to ensure sanctions are properly enforced. Where this integration is weak or absent, institutions report cases of students continuing to access academic services despite active sanctions, underscoring the need for systems where enforcement is not a manual, easily-missed step but a built-in part of the registration workflow.

Transparency and Appeals in Institutional Processes

Research on procedural fairness in institutional decision-making emphasizes that transparency and a clear appeals mechanism significantly increase trust in the outcome of a process, even when the outcome itself is unfavorable to the individual. Systems that allow individuals to track the status of their case and formally contest a decision tend to be perceived as fairer than opaque, purely administrative processes.

Identification of Gaps in Existing Literature

7. Limited adaptation to local institutional context: Most existing student conduct systems are built for specific regulatory environments and are not readily adaptable to universities in different regions, such as AUCA.
8. Weak integration with registration systems: Many case management tools handle disciplinary records in isolation, without automatically enforcing sanctions through registration status.
9. Lack of structured re-integration tracking: Few systems formally verify and record a student's return to normal standing after a sanction, leaving this step informal and undocumented.
10. Limited transparency for students: Existing manual processes rarely give students real-time visibility into the status of their own case.
11. High cost and complexity of commercial solutions: Proprietary student conduct platforms are often expensive and over-engineered for the needs of a single institution.

Summary

The reviewed literature confirms that digital case management consistently improves the speed, consistency, and transparency of administrative processes, and that these benefits extend naturally to student disciplinary management. However, existing solutions are either too generic, too costly, or insufficiently integrated with academic registration systems to be directly suitable for AUCA. CaseFlow addresses these gaps by providing a purpose-built, institution-specific system that directly ties enforcement to registration status, formally tracks re-integration, and gives students clear visibility into their case — closing the gap between general case management theory and the specific needs of university disciplinary processes.

PROPOSED SOLUTION

Overview of the Solution

CaseFlow is a web-based platform that digitizes the entire student disciplinary case lifecycle at AUCA. The system replaces fragmented, paper-based handling with a single, centralized workflow covering incident reporting, committee review, enforcement, status tracking, appeals, and re-integration verification. Role-based access ensures that students, reporting staff, committee members, and administrators each interact with the system according to their responsibilities, while a complete audit trail preserves the integrity and accountability of every case.

By tying enforcement directly to a student’s registration status, CaseFlow ensures that sanctions are consistently applied without relying on manual cross-checking between offices. By formally recording re-integration, the system gives both students and staff a clear, verifiable confirmation that a student has returned to good standing, reducing disputes and confusion at the end of a disciplinary process.

System Design

High-Level System Architecture

CaseFlow follows a layered architecture comprising a web-based front-end for students, staff, and committee members; a back-end application server that manages business logic, authentication, and role-based permissions; and a relational database that stores case records, user accounts, decisions, and the full audit history of each case. A notification component sends automated alerts at key transition points in a case’s lifecycle, and an integration layer connects disciplinary enforcement status with the university’s student registration data.

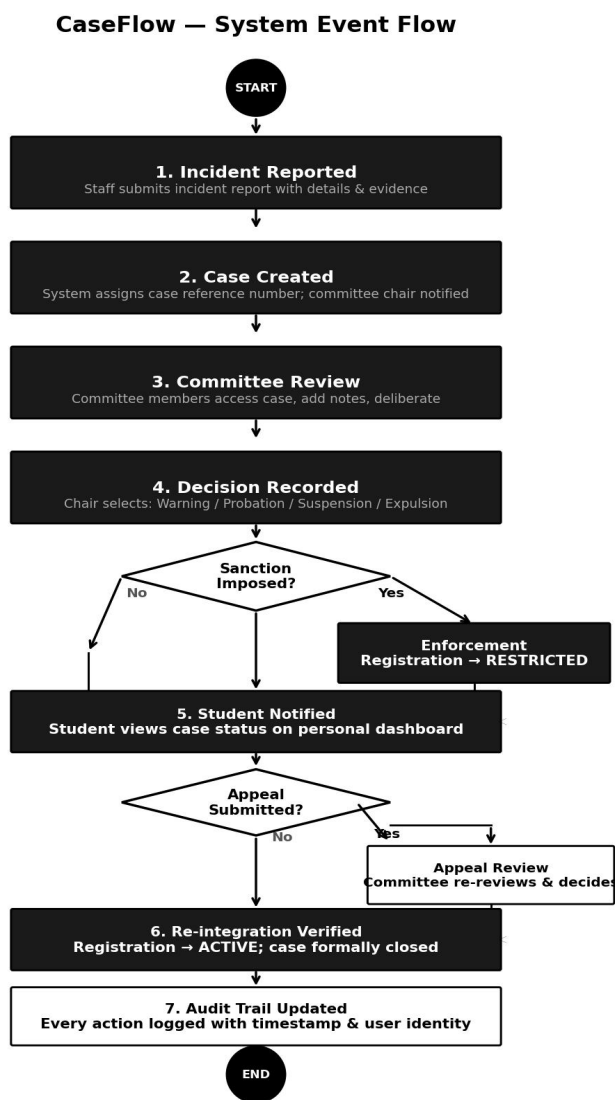


Figure 1: CaseFlow System Architecture — Incident Reporting → Committee Review → Decision → Enforcement → Appeal (if any) → Re-integration

Key Features of the Proposed Solution

- Incident Reporting Module: Secure, structured submission of disciplinary incidents by authorized staff, including category, description, and supporting evidence.
- Committee Review Workspace: A shared digital space where committee members review case details, add deliberation notes, and record formal decisions.
- Enforcement Engine: Automatic linkage between a recorded decision and the student’s registration status, restricting or flagging access as required.
- Status Tracking Dashboard: Real-time visibility of a case’s current stage for students and authorized staff.
- Appeals Workflow: A structured process for students to formally contest a decision, with committee re-review and resolution tracking.
- Re-integration Verification: Formal confirmation and recording of a student’s return to normal academic standing after completing a sanction.
- Role-Based Access Control: Distinct permission levels for students, staff, committee members, and administrators.
- Audit Trail and Case History: A complete, timestamped log of every action taken on a case.

- Automated Notifications: Email or in-app alerts at key stages, such as case opened, decision made, appeal submitted, or re-integration confirmed.

METHODOLOGY

This project will follow the Agile Software Development methodology, specifically the Scrum framework, due to its flexibility and suitability for projects with evolving requirements. Disciplinary processes involve nuanced rules and stakeholder input from the committee, registration office, and student affairs office, all of which benefit from iterative refinement through short development sprints, regular feedback, and incremental delivery of features such as incident reporting, committee review, enforcement, and re-integration verification.

Agile is appropriate for CaseFlow because several aspects of the system — particularly the enforcement integration with registration and the exact structure of the appeals workflow — may require adjustment as the project team gathers more detailed input from university stakeholders. Sprint reviews and retrospectives will allow issues to be identified and corrected early, reducing the risk of building a system that does not match real institutional processes.

Tools and Technologies

Programming Languages

- JavaScript / TypeScript – For front-end and back-end development.
- SQL – For database design and queries.

Frameworks and Libraries

- React – For building the web-based user interface.
- Node.js with Express, or Django REST Framework – For back-end API development.

Databases

- PostgreSQL or MySQL – Primary relational database for case records, user accounts, and audit logs.

Design & Collaboration Tools

- Figma – For UI/UX design and prototyping.
- GitHub – For version control and collaboration.

Data Collection

CaseFlow will require structured information about the existing disciplinary process, including the typical stages a case moves through, the roles and responsibilities of committee members, the criteria used for decisions and sanctions, and the current re-integration procedure. This information will be gathered through interviews with student affairs staff and disciplinary committee members, review of existing (paper-based) case records and policy documents, and consultation with the registration office regarding how sanctions should be reflected in student registration status.

Testing and Validation

Testing and Strategy

- Unit Testing: To validate individual components such as form validation, authentication, and status update logic.
- Integration Testing: To ensure correct communication between the reporting, review, enforcement, and registration-linked modules.
- User Acceptance Testing (UAT): Conducted with sample staff and committee members to confirm the system matches real institutional workflows.

Validation of Results

- Verification that enforcement actions correctly restrict registration status when a sanction is active.
- Confirmation that case status updates are accurately reflected in real time for students and staff.
- Review of audit logs for completeness and accuracy across the full case lifecycle.

WORK PLAN AND TIMELINE

The project will be organized using a Work Breakdown Structure (WBS) guided by SMART criteria, dividing the work into clearly defined phases with deliverables and milestones.

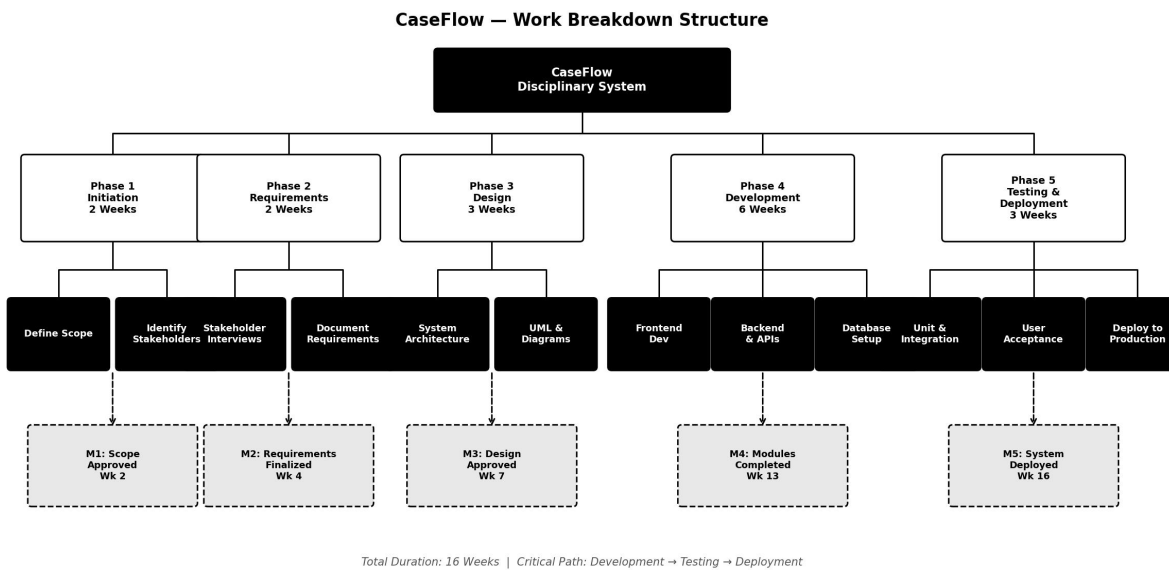


Figure 2: CaseFlow Work Breakdown Structure

1. Project Initiation

Purpose: Establish the foundation for the project.

- Define overall project scope (goals, deliverables, constraints).
- Identify stakeholders, including student affairs, the disciplinary committee, and the registration office.

Milestone: Scope Document Approved.

2. Requirements Gathering

Purpose: Capture and validate all user, system, and process requirements.

- Conduct interviews with disciplinary committee members and student affairs staff.
- Document functional and non-functional requirements.
- Define enforcement and registration-integration requirements.

Milestone: Requirements Document Finalized.

3. System Design

Purpose: Plan the full system architecture and workflow logic.

- Design system architecture (front-end, back-end, database schema).
- Define case lifecycle workflow (reporting → review → decision → enforcement → appeal → re-integration).
- Create UML diagrams and data flow diagrams.

Milestone: Design Approved.

4. Development Phase

Purpose: Build and integrate the system components.

- Front-End Development: reporting forms, committee dashboard, student status view.
- Back-End Development: authentication, case engine, enforcement logic, notifications.
- Database Implementation: case records, audit trail, user roles.

Milestone: System Modules Completed.

5. Testing and Deployment

Purpose: Validate and deploy the system.

- Unit, integration, and user acceptance testing.
- Bug fixing and refinement based on UAT feedback.
- Deployment to a test/production environment.

Milestone: System Deployed.

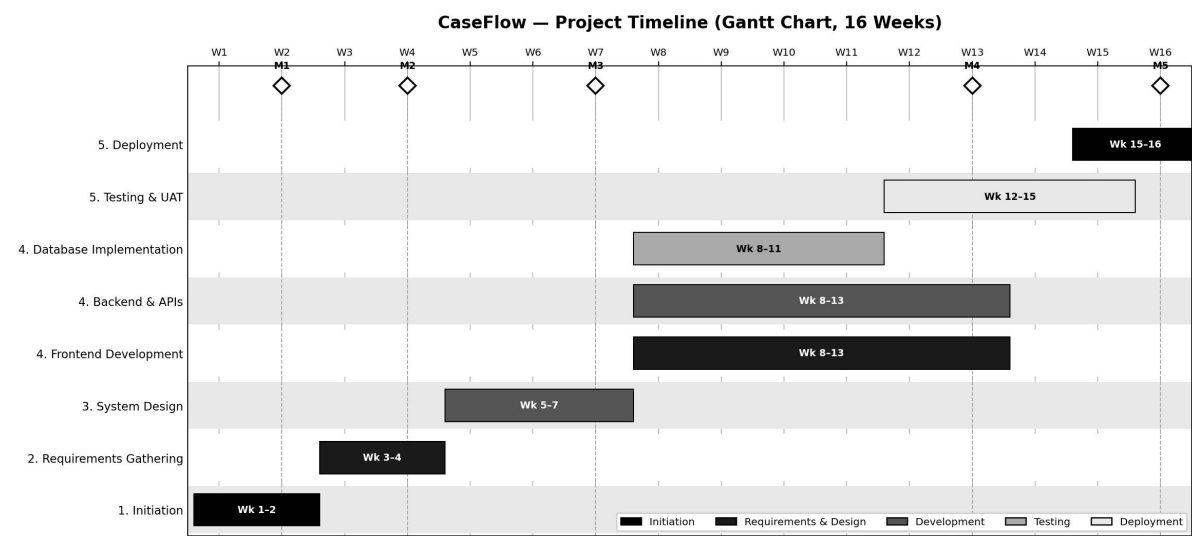


Figure 3: Gantt Chart — Estimated 16-Week Timeline (Initiation: Weeks 1–2; Requirements: Weeks 3–4; Design: Weeks 5–7; Development: Weeks 8–13; Testing & Deployment: Weeks 14–16)

EXPECTED OUTCOMES

Expected Deliverables

12. Fully Functional System: A complete, operational CaseFlow application implementing incident reporting, committee review, enforcement, status tracking, appeals, and re-integration verification.
13. Database Implementation and Schema: A fully designed and implemented database storing case records, user roles, and audit logs.
14. System Documentation: Architecture documentation, user manual, installation guide, and developer guide.
15. Testing and Validation Reports: Test cases, results, and UAT feedback summaries.
16. Project Presentation and Final Report: An academic report and presentation summarizing objectives, methodology, implementation, and results.

Potential Limitations

- Adoption and Change Management: Staff and committee members accustomed to manual processes may require training and time to adapt to the new system.
- Data Sensitivity: Disciplinary records are highly sensitive and will require careful access control and data protection measures.
- Integration Constraints: Linking enforcement to registration status depends on access to, and cooperation from, the university's existing registration system.
- Internet Dependency: As a web-based system, reliable functioning depends on stable campus network connectivity.
- Policy Alignment: The system must be carefully aligned with existing disciplinary policy, which may require ongoing consultation with student affairs throughout development.

CONCLUSION

CaseFlow addresses a real and persistent gap in how student disciplinary matters are currently handled at AUCA — a process that remains largely manual despite the sensitivity and importance of fair, timely, and transparent handling of student conduct cases. By digitizing the full case lifecycle, from incident reporting through committee review, enforcement, appeals, and re-integration, the system aims to reduce delays, improve transparency for students, ensure consistent enforcement of decisions, and maintain a reliable, auditable record of every case.

The project will be developed using an Agile methodology over an estimated 16-week period, with close consultation with student affairs staff and the disciplinary committee to ensure the system reflects real institutional processes and policy requirements. Upon completion, CaseFlow is expected to deliver a fully functional, role-based case management system that strengthens trust in the disciplinary process while reducing the administrative burden on staff and committee members. The calm, structured approach embodied in the name “CaseFlow” reflects the project’s broader goal: to make a sensitive process feel orderly, fair, and respectful for every student involved.

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APPENDICES

Data Collection Letter

[Attach a signed letter requesting permission to collect data and conduct interviews with student affairs staff and disciplinary committee members for the purpose of this project.]

Case Study Organization Approval Letter

[Attach a signed approval letter from AUCA authorizing CaseFlow as the approved capstone case study/project topic.]